UNIVERSITÉ DE NAMUR

UNamur Robotics Lab
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MAKER FAIRE ROME 2026

Operator Manual

Setting up and running the e-puck2 swarm robot stand: six robots that visitors drive from their own phones.

6 robots · 1 router · 1 laptop · about 10 minutes to set up

IN THIS MANUAL

1 The kit and the network — page 2

2 Setting up, steps 1–4 — page 3

3 All robots, during the fair, end of day — page 4

4 Turns and the operator page — page 5

5 Troubleshooting and quick reference — page 6

6 Robot QR codes — page 7

All files on GitHub

Robot software, poster, QR labels and troubleshooting notes. Scan the code or open:

`github.com/unamurroboticslab-hue/maker-faire-rome-2026`

1 The kit and the network

WiFi router

Pre-configured. Plug in and switch on. No internet needed.

6 e-puck2 robots

Software already installed in /home/pi/blocky. Numbered labels on top.

Your laptop

Any laptop with a terminal. Used to start each robot's server.

Robot WiFi

NETWORK	RoboticsLab
PASSWORD	EpuckRobot

Robots and addresses

ROBOT	IP ADDRESS	CONNECT (LAPTOP)	REMOTE CONTROL (PHONE)
#1	192.168.0.111	ssh pi@192.168.0.111	.111:5000/mobile.html
#2	192.168.0.113	ssh pi@192.168.0.113	.113:5000/mobile.html
#3	192.168.0.114	ssh pi@192.168.0.114	.114:5000/mobile.html
#4	192.168.0.115	ssh pi@192.168.0.115	.115:5000/mobile.html
#6	192.168.0.118	ssh pi@192.168.0.118	.118:5000/mobile.html
#12	192.168.0.112	ssh pi@192.168.0.112	.112:5000/mobile.html

Full phone address: http://192.168.0.1xx:5000/mobile.html. Block Coder runs at http://<IP>:5000/

⚠ **Never factory-reset the router.** The robot addresses are reserved in it. After a reset the addresses change and none of the QR codes will work.

2 Setting up

Do steps 1–4 for the first robot. Page 4 covers the other five.

1 Switch on the router

Plug it in and wait about **30 seconds**. Then connect your laptop to the WiFi **RoboticsLab** (password **EpuckRobot**). The laptop and all robots must be on this network.

2 Switch on a robot

Flip the power switch on the side. Wait **30–40 seconds** for it to boot and join the WiFi. The LEDs flicker while it boots; wait until they settle.

3 Connect to the robot from your laptop

Open a terminal (Linux, Mac) or PowerShell (Windows) and type, with the robot's address from page 2:

```
$ ssh pi@192.168.0.111
```

PASSWORD **11110000** same for all robots

Nothing appears on screen while you type the password; that's normal. The first time, SSH asks *"Are you sure you want to continue connecting?"* Type **yes** and press Enter.

4 Start the robot's server

Once connected, type these two commands, pressing Enter after each:

```
pi@epuck2:~ $ cd /home/pi/blocky
pi@epuck2:~/blocky $ python3 server.py
```

The server prints its startup messages and keeps running: the robot is ready. **Leave this terminal window open**. Closing it stops the robot.

3 All robots, the fair, end of day

5 Repeat for every robot

Open a **new terminal window** for each robot and do steps 2–4 with its address. At the end you have six windows, each running `python3 server.py`. Keep the laptop plugged in and stop it from going to sleep.

6 Test each robot with your phone

Join **RoboticsLab** on your phone, scan each robot's QR code on the poster, tap **Connetti** and drive it a little. If one doesn't respond, see page 6 before visitors arrive.

During the fair

VISITOR STEP 1

Scan the WiFi code on the poster to join **RoboticsLab**.

VISITOR STEP 2

Scan the code of the robot with the matching number, then tap **Connetti**.

IF IT WON'T OPEN

Phone says "no internet": choose **Stay connected**, turn off mobile data. Address starts with https? Change it to **http**.

- One phone drives each robot at a time. To give the next child a turn, use **End turn** on the operator page (page 5).
- Check the robots' batteries every hour or so. A robot that slows down or stops answering usually needs charging.
- Keep the six terminal windows open on the laptop all day.

End of the day

Switch each robot off with its two **blue buttons**, in this order:

1. Press the **blue button on the robot** (the e-puck2) to switch it off.
2. Press and **hold the blue button on top of the Pi-puck for 6–7 seconds**, until it switches off.

You don't need to type anything in the terminal. Switch off the router last. Put the robots on charge for the next day.

4 Turns and the operator page

Only **one phone drives each robot at a time**. The first phone that taps **Connetti** drives. Other phones see *"Robot occupato — aspetta il tuo turno!"* and their arrows stay grey. When the driving phone disconnects or closes the page, the next waiting phone gets the robot by itself.

Before the fair, once: update the robots

1. Get the files (needs internet): open `github.com/unamurroboticslab-hue/maker-faire-rome-2026`, click **Code › Download ZIP** and unzip it.

2. Copy them to each robot. Connect the laptop to RoboticsLab, open a terminal in the unzipped folder and run this once per robot, with that robot's address in place of 1xx (password **11110000**):

```
$ cd robot/blocky
$ scp mobile.html operator.html server.py pia192.168.0.1xx:/home/pi/blocky/
```

3. Restart each robot's server: in its terminal window press **Ctrl + C**, then type `python3 server.py`. A robot whose server isn't running yet just needs the usual steps 2–4 on page 3.

Open the operator page on the laptop

ADDRESS (ANY ROBOT)
http://192.168.0.111:5000/operator.html

OPERATOR KEY
11110000

One card per robot shows **which phone is driving and for how long**, the other phones waiting, and the blocked phones. Phones are shown by their WiFi address. The page refreshes every 2 seconds.

BUTTON	WHAT IT DOES
End turn	Stops the robot and frees it for the next child. That phone sees "Turno finito!" and must wait before driving again. Choose the wait (1, 2, 5 or 10 min) at the top of the page.
Block	Keeps a phone out until you press Unblock. Works on the driving phone and on waiting phones.
Unblock	Lets a blocked phone play again.
Free robot	Clears the driver and every block on that robot.

OFFLINE
The robot is off, or its server isn't running.

WRONG KEY
Type 11110000 in the Operator key box.

OLD SERVER.PY
That robot wasn't updated. Copy the files (box above) and restart it.

5 Troubleshooting

PROBLEM	WHAT TO DO
SSH: connection refused, or no route to host	The robot is still booting. Wait 30 s and try again. Check the laptop is on RoboticsLab.
SSH: permission denied	Retype the password **11110000** slowly. Nothing shows while you type.
SSH: "remote host identification has changed"	Run `ssh-keygen -R 192.168.0.1xx` with the robot's address, then connect again.
"Address already in use" when starting the server	An old server is still running. Run `pkill -f server.py`, then start it again.
Server stopped with an error	Run `python3 server.py` again in the same window.
Phone: page doesn't open	The phone must be on RoboticsLab. Turn off mobile data. On "no internet", choose Stay connected. "Can't provide a secure connection": change **https** to **http** in the address.
Phone: "Robot occupato" or grey arrows	Another phone is driving, or this phone's turn was ended. Tap Connetti, or use End turn / Unblock on the operator page (page 5).
Page opens but the robot doesn't move	Check that robot's terminal window is still running the server. Then check its battery.
Robot never joins the network	Switch it off and on, wait 40 s. Make sure the router from the kit is the one switched on.
Robot files damaged or missing	Copy them back from the GitHub repo: `scp -r robot/blocky pi@<IP>:/home/pi/`

Quick reference

cut out and keep by the laptop

START A ROBOT

```
ssh pi@192.168.0.1xx
password 11110000
cd /home/pi/blocky
python3 server.py
```

STOP A ROBOT

1. Blue button on the robot
2. Hold Pi-puck top button 6–7 s

WIFI

RoboticsLab · EpuckRobot

ROBOT ADDRESSES

#1 .111 #2 .113 #3 .114

#4 .115 #6 .118 #12 .112

all start with 192.168.0.

OPERATOR PAGE

.111:5000/operator.html

6 Robot QR codes

Spare codes for testing, or to cut out along the dashed lines and stick on top of each robot. Join the WiFi first, then scan a robot to open its remote control.

1 Join the robot WiFi

Network **RoboticsLab** · Password **EpuckRobot**

No internet on this network. Choose "Stay connected" if the phone asks.

2 Scan a robot

Robot 1 e-puck2

192.168.0.111

Robot 2 e-puck2

192.168.0.113

Robot 3 e-puck2

192.168.0.114

Robot 4 e-puck2

192.168.0.115

Robot 6 e-puck2

192.168.0.118

Robot 12 e-puck2

192.168.0.112

Robots 12, 16, 24 and 25

Join the WiFi first, then scan a robot to open its remote control. Cut along the dashed lines to stick a code on each robot.

1 Join the robot WiFi

Network **RoboticsLab** · Password **EpuckRobot**

No internet on this network. Choose "Stay connected" if the phone asks.

2 Scan a robot

Robot 12

e-puck2

192.168.0.112

Robot 16

e-puck2

192.168.0.116

Robot 24

e-puck2

192.168.0.124

Robot 25

e-puck2

192.168.0.125